

Syeda Anika Jerin

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EDUCATION

- **Jahangirnagar University, Savar**

Savar, Dhaka

B.Sc. in Information Technology; CGPA: 3.48 out of 4.00

Feb. 2015 – Dec. 2018

PROFESSIONAL SUMMARY

Software Engineer with 5+ years of experience leading the planning, design, and delivery of large-scale, AI-driven software systems. Proven in taking end-to-end ownership of projects, aligning technical strategy with business objectives, and delivering high-impact solutions for government and enterprise clients. Experienced in mentoring engineers, conducting technical reviews, and driving a culture of quality, accountability, and continuous improvement while leveraging AI and emerging technologies to enhance efficiency and delivery.

EXPERIENCE

- **Unisoft Systems Ltd, a sister concern of Smart Technologies (BD)**

Dhaka, Bangladesh

Software Engineer

Dec 2020 – Oct 2025

- **AI-Driven Traffic Monitoring System:** Contributed to the development of an AI-driven traffic monitoring and auto-fining system for the Bangladesh Highway Police as part of a government-funded project with an overall budget of approximately 72 crore BDT, supporting large-scale deployment and operational use. Developed and deployed system components to detect traffic violations such as speeding and wrong-way driving, including automated number plate recognition. Fine-tuned and optimized deep learning models and built backend services to support real-world inference workflows. Performed exploratory data analysis, data augmentation, and model evaluation on road traffic datasets specific to Bangladesh.
- **Weather Forecast Production and Archiving System:** Led the development of core components of a web-based meteorological forecasting and archiving platform for the Bangladesh Meteorological Department under a project funded by the Government of Bangladesh and the World Meteorological Organization (WMO), with an overall budget of approximately 30 crore BDT. Designed and implemented the system to collect, decode, process, and store diverse meteorological data formats (SYNOP, METAR, TAF, BUFR, etc.). Built data analysis pipelines and interactive visualizations to support operational forecasting and long-term data archiving.
- **AI-Driven Quality Monitoring Module (ERP Integration):** Customized existing modules and developed new modules integrated with sales, purchase, inventory, billing, accounting, manufacturing, warehouse in python (Odoo Framework). Developed and integrated AI-based quality monitoring components within an ERP system, experimenting with machine learning models to analyze production data and textual feedback for defect prediction and process insights. Ensured alignment with operational and quality control requirements.
- **Education Management System:** Designing, building, and maintaining efficient, reusable, and reliable code. Co-operate with the developer team to automate the admission process, record management, fee collection, attendance process, and online examination for the Smart Academy School. Demonstrated SOLID Design Principles, source control, build, bug-tracking system, and coding standards.
- **Backend APIs and System Integration:** Designed and implemented RESTful backend APIs to serve machine learning models and support internal web applications, contributing to scalable system integration and reliable application workflows.
- **Tech Stack:** Python, Odoo, PyTorch, TensorFlow, Flask, Django, PostgreSQL, NumPy, Pandas, OpenCV, Plotly, Folium, Git etc.
- **Machine Learning & Models:** LSTM, CNNs, Vision Transformers, YOLOv8, YOLO-NAS, Deep SORT, Random Forest, XGBoost, Isolation Forest, One-Class SVM

- **Smarter Stock – Ecosoftbd**

Remote

Software Engineer

September 2020 – November 2020

- **Stock Prediction:** Manipulated large data sets and used them to identify trends and reached meaningful conclusions to inform strategic business decisions (Stock market). Cleaned, aggregated, and organized data from disparate sources. Applied ML algorithm (LSTM) on processed datasets.

PROJECTS

- **Cross-Modal-DL-Framework-for-Alzheimer-s-Disease-Classification-Using-MRI-Data-and-Tabular-Features (Research Project):** Implemented a cross-modal deep learning framework that fuses MRI imaging data with tabular clinical features for Alzheimer's disease classification. It provides the core model architecture, data preprocessing pipeline, and training routines as a reference (v1). The approach demonstrates how multimodal integration can improve classification over single-modality baselines.
- **Quantum-Enhanced-Graph-Learning-for-Molecular-Design-in-Biomaterials (Research Project):** A hybrid classical-quantum pipeline where a GNN encoder learns molecular structure features and a quantum variational head predicts (1) electronic bandgap (regression; we'll use HOMO-LUMO gap as a proxy) and (2) toxicity risk (classification). The system supports conditional generation hooks later, but the MVP focuses on predictive modeling.
- **ChaGPT-Extension-for-Youtube (Academic Project):** This Chrome extension enables users to ask questions about YouTube videos in real time by extracting the video's transcript, sending it to ChatGPT, and displaying responses directly on the YouTube page through a custom chat interface. It provides an interactive and AI-enhanced experience for better understanding video content.
- **Face-Recognition-OpenCV (Academic Project):** A real-time face recognition system built with Python, OpenCV, and Flask that integrates with Odoo employee attendance. It detects and recognizes faces via webcam, then interacts with Odoo to automate tasks like attendance tracking. The project combines computer vision with business workflow automation for seamless identity verification.
- **AI-Desktop-Assistant (Academic Project):** The AI Desktop Assistant is a Python-based desktop application that utilizes speech recognition to perform various tasks. Users can command the assistant to open websites like Google, YouTube, GitHub, and Stack Overflow, play music from their desktop, and even send emails. This project serves as an interactive and fun way to automate routine computer operations through voice commands.
- **Python library-geomap3D (Academic Project):** GeoMap3D is a Python library that makes creating beautiful, interactive 3D geographic mapping, without requiring any JavaScript, frontend frameworks, or GIS software. GeoMap3D lets you generate and export rich 3D graphs using a clean, declarative Python API.

CERTIFICATION/AWARDS

- **Calculus for Machine Learning and Data Science:** Issuing Organization: DeepLearning.AI and Credential ID: XFEZ0YSMA3GV
- **Advanced Computer Vision with TensorFlow:** Issuing Organization: DeepLearning.AI — and Credential ID: LLNLAFN5NSBE
- **Government Scholarship:** Recipient of the Government scholarship for outstanding results in the Higher Secondary Certificate (H.S.C), Secondary School Certificate (S.S.C), and Junior School Certificate (J.S.C) examinations. Awarded by the Intermediate and Secondary Education Board, Bangladesh.
- **Best Paper Award:** Best Impact Award received for the co-authored conference paper, COVID-Hero: Machine Learning Based COVID-19 Awareness Enhancement Mobile Game for children.

SKILLS

- **IDEs:** Codeblocks, Pycharm, Visual Studio Code, Jupyter, Postman.
- **Editors:** Notepad++, Sublime Text, Microsoft Office
- **Database Tools:** Pgadmin, phpMyAdmin
- **Open Source Library:** Pytorch, OpenCV, Keras, Tensorflow, Numpy, Pandas, Scikit learn, Matplotlib, Plotly, Folium, Leaflet JS, React Js etc.
- **Database:** PostgreSQL, MySQL
- **Programming Language:** Python, C, C++, MATLAB, Golang
- **API:** RestAPI, FastAPI
- **Version Control:** Git, Github
- **Web Technologies:** HTML, CSS, JavaScript, SQL, XML
- **Frameworks:** Odoo, Flask, Django **OS:** Windows, Linux, Mac